

ISRO 2025: Topic-Wise Practice Questions

TOPIC 6: COMPUTER ORGANIZATION & ARCHITECTURE

Based on 2023 ISRO Questions: 17, 18, 19, 20, 21, 22, 23

****Coverage Analysis****: This topic had 7 questions (8.75%) in 2023 paper, focusing on processor architectures, pipeline implementation, instruction execution, and arithmetic operations.

****Key Concepts Tested****: Expression evaluation, vectored interrupts, processor architectures, pipeline implementation, micro-operations, error detection methods.

INSTRUCTION SET ARCHITECTURE (Questions 1-15)

****1. In Reverse Polish notation, expression $P*Q+R*S$ is written as:****

- (a) $PQ*RS*+$
- (b) $P*QRS*+$
- (c) $PQ*RS+*$
- (d) $P*Q*RS+$

****2. Minimum arithmetic operations to evaluate $f(x,y,z) = x^3 + 2x^2y + 3xy^2 + y^3$:****

- (a) 6
- (b) 8
- (c) 9
- (d) 10

****3. Prefix expression evaluation $-+abc*de$ with $a=2, b=3, c=1, d=4, e=5$:****

- (a) 14
- (b) 16
- (c) 18
- (d) 20

****4. CISC processors have:****

- (a) Complex instructions, fewer registers
- (b) Simple instructions, more registers
- (c) Fixed-length instructions
- (d) No memory-to-memory operations

****5. RISC processors have:****

- (a) Simple instructions, more registers
- (b) Complex instructions, fewer registers
- (c) Variable-length instructions
- (d) Complex addressing modes

****6. Load-store architecture means:****

- (a) Only load/store instructions access memory

- (b) All instructions can access memory
- (c) No register operations
- (d) Memory-to-memory operations allowed

****7. Addressing modes include:****

- (a) Direct, indirect, immediate
- (b) Register, indexed, relative
- (c) Base+displacement, auto-increment
- (d) All of above

****8. Instruction format typically contains:****

- (a) Opcode and operands
- (b) Addressing mode bits
- (c) Condition code bits
- (d) All of above

****9. Zero-address instruction uses:****

- (a) Stack for operands
- (b) Accumulator for operands
- (c) Memory for operands
- (d) Registers for operands

****10. Three-address instruction format:****

- (a) ADD R1, R2, R3 ($R1 = R2 + R3$)
- (b) ADD R1, R2 ($R1 = R1 + R2$)
- (c) ADD (ACC = ACC + operand)
- (d) ADD (stack operation)

****11. Instruction cycle phases:****

- (a) Fetch, Decode, Execute
- (b) Fetch, Decode, Execute, Store
- (c) Fetch, Execute
- (d) Decode, Execute

****12. Program counter (PC) contains:****

- (a) Address of current instruction
- (b) Address of next instruction
- (c) Address of previous instruction
- (d) Address of data

****13. Instruction register (IR) contains:****

- (a) Current instruction being executed
- (b) Next instruction to execute
- (c) Address of instruction
- (d) Result of instruction

****14. Condition codes are set by:****

- (a) Arithmetic operations

- (b) Logical operations
- (c) Compare operations
- (d) All of above

****15. Branch instructions use:****

- (a) Condition codes for conditional branches
- (b) Direct addressing for unconditional branches
- (c) Relative addressing often
- (d) All of above

**PROCESSOR ARCHITECTURE & DESIGN (Questions 16-30)**

****16. For given finite instructions, which processor architecture provides fastest execution:****

- (a) VLIW (Very Long Instruction Word)
- (b) Super-scalar
- (c) RISC
- (d) CISC

****17. Super-scalar processors can:****

- (a) Execute multiple instructions per clock cycle
- (b) Issue multiple instructions simultaneously
- (c) Have multiple execution units
- (d) All of above

****18. VLIW processors:****

- (a) Pack multiple operations in single instruction
- (b) Rely on compiler for instruction scheduling
- (c) Have fixed instruction format
- (d) All of above

****19. Out-of-order execution:****

- (a) Executes instructions as they become ready
- (b) Maintains program semantics
- (c) Improves performance by avoiding stalls
- (d) All of above

****20. Register renaming eliminates:****

- (a) True dependencies
- (b) Anti-dependencies (WAR)
- (c) Output dependencies (WAW)
- (d) Both (b) and (c)

****21. Branch prediction improves performance by:****

- (a) Predicting branch outcomes
- (b) Reducing branch penalties
- (c) Enabling speculative execution
- (d) All of above

****22. Dynamic branch prediction uses:****

- (a) Branch history
- (b) Pattern recognition
- (c) Statistical prediction
- (d) All of above

****23. Speculative execution:****

- (a) Executes instructions before branch resolution
- (b) Rolls back on misprediction
- (c) Improves performance
- (d) All of above

****24. Data hazards in pipeline occur due to:****

- (a) Data dependencies between instructions
- (b) Resource conflicts
- (c) Control flow changes
- (d) Memory access patterns

****25. Forwarding (bypassing) reduces:****

- (a) Data hazard penalties
- (b) Control hazard penalties
- (c) Structural hazard penalties
- (d) All hazards

****26. Multi-core processors have:****

- (a) Multiple CPU cores on single chip
- (b) Shared cache hierarchy
- (c) Independent execution units
- (d) All of above

****27. Cache coherence protocols ensure:****

- (a) Consistency of shared data
- (b) Performance optimization
- (c) Memory hierarchy management
- (d) All of above

****28. Simultaneous Multi-Threading (SMT):****

- (a) Executes multiple threads on single core
- (b) Shares execution units between threads
- (c) Also called Hyper-Threading
- (d) All of above

****29. Vector processors excel at:****

- (a) Scientific computations
- (b) Multimedia processing
- (c) Parallel data operations
- (d) All of above

****30. GPU architecture is optimized for:****

- (a) Highly parallel computations
- (b) Graphics rendering
- (c) SIMD operations
- (d) All of above

**PIPELINE PROCESSING (Questions 31-40)**

****31. An instruction pipeline can be implemented by:****

- (a) LIFO buffer
- (b) FIFO buffer
- (c) Stack
- (d) Queue

****32. 5-stage pipeline consists of:****

- (a) IF, ID, EX, MEM, WB
- (b) Fetch, Decode, Execute, Store, Commit
- (c) IF, OF, EX, WB, ST
- (d) All are valid representations

****33. Pipeline speedup is limited by:****

- (a) Pipeline depth
- (b) Hazards and stalls
- (c) Branch frequency
- (d) All of above

****34. Structural hazards occur when:****

- (a) Hardware resources are insufficient
- (b) Data dependencies exist
- (c) Control flow changes
- (d) Instructions conflict

****35. Control hazards are caused by:****

- (a) Branch instructions
- (b) Data dependencies
- (c) Resource conflicts
- (d) Memory access

****36. Pipeline stall occurs when:****

- (a) Hazard cannot be resolved
- (b) Resources are unavailable
- (c) Branch misprediction
- (d) All of above

****37. Delayed branch technique:****

- (a) Places useful instruction after branch
- (b) Reduces branch penalty
- (c) Requires compiler optimization

(d) All of above

****38. Loop unrolling:****

- (a) Reduces loop overhead
- (b) Increases instruction-level parallelism
- (c) May increase code size
- (d) All of above

****39. Pipeline efficiency formula:****

- (a) Actual CPI / Ideal CPI
- (b) Instructions executed / Clock cycles
- (c) Speedup / Pipeline depth
- (d) Throughput / Latency

****40. Deep pipelines have:****

- (a) Higher clock frequency potential
- (b) Higher branch penalty
- (c) More complex hazard handling
- (d) All of above

**MEMORY HIERARCHY & ARCHITECTURE (Questions 41-50)**

****41. Memory hierarchy is organized by:****

- (a) Speed, cost, capacity
- (b) Temporal and spatial locality
- (c) Access patterns
- (d) All of above

****42. Cache memory exploits:****

- (a) Temporal locality
- (b) Spatial locality
- (c) Both temporal and spatial locality
- (d) Random access patterns

****43. Cache hit ratio formula:****

- (a) Hits / (Hits + Misses)
- (b) Misses / Total accesses
- (c) Access time / Miss penalty
- (d) Cache size / Memory size

****44. Direct-mapped cache:****

- (a) Each memory block maps to exactly one cache line
- (b) Simple implementation
- (c) May have high conflict misses
- (d) All of above

****45. Associative cache:****

- (a) Memory block can map to any cache line

- (b) Requires content addressable memory
- (c) Expensive implementation
- (d) All of above

****46. Set-associative cache:****

- (a) Combines direct-mapped and associative
- (b) Memory block maps to specific set
- (c) Within set, can go to any line
- (d) All of above

****47. Cache replacement policies:****

- (a) LRU (Least Recently Used)
- (b) FIFO (First In First Out)
- (c) Random replacement
- (d) All of above

****48. Write-through cache:****

- (a) Writes to cache and memory simultaneously
- (b) Ensures memory consistency
- (c) Higher write latency
- (d) All of above

****49. Write-back cache:****

- (a) Writes only to cache initially
- (b) Writes to memory on replacement
- (c) Uses dirty bit
- (d) All of above

****50. Virtual memory provides:****

- (a) Larger address space than physical memory
- (b) Memory protection between processes
- (c) Automatic memory management
- (d) All of above

**ARITHMETIC & LOGIC UNIT (Questions 51-55)**

****51. Operation executed on data stored in registers is called:****

- (a) Macro-operation
- (b) Micro-operation
- (c) Bit-operation
- (d) Word-operation

****52. Which method detects double errors:****

- (a) Odd parity
- (b) Even parity
- (c) Hamming code
- (d) CRC (Cyclic Redundancy Check)

****53. IEEE 754 floating-point standard uses:****

- (a) Sign bit, exponent, mantissa
- (b) Biased exponent representation
- (c) Normalized mantissa
- (d) All of above

****54. Booth's algorithm is used for:****

- (a) Integer multiplication
- (b) Floating-point addition
- (c) Division
- (d) Square root

****55. In a vectored interrupt:****

- (a) Branch address assigned to fixed location
- (b) Interrupting source supplies branch information
- (c) Branch address from processor register
- (d) Software determines branch address

**ANSWER KEY - COMPUTER ORGANIZATION & ARCHITECTURE**

****1-10:**** (a), (c), (b), (a), (a), (a), (d), (d), (a), (a)

****11-20:**** (a), (b), (a), (d), (d), (b), (d), (d), (d), (d)

****21-30:**** (d), (d), (d), (a), (a), (d), (a), (d), (d), (d)

****31-40:**** (b), (a), (d), (a), (a), (d), (d), (d), (a), (d)

****41-50:**** (d), (c), (a), (d), (d), (d), (d), (d), (d), (d)

****51-55:**** (b), (d), (d), (a), (b)

**Critical Study Focus for September 14th:**

1. ****Instruction Processing (Q1-15)**:** Master expression evaluation, addressing modes, instruction formats
2. ****Processor Design (Q16-30)**:** Understand different architectures (RISC, CISC, super-scalar, VLIW)
3. ****Pipeline Concepts (Q31-40)**:** Focus on hazards, stalls, and performance optimization techniques
4. ****Memory Systems (Q41-50)**:** Know cache organization, replacement policies, and virtual memory
5. ****Arithmetic Operations (Q51-55)**:** Understand micro-operations, error detection, and specialized algorithms

****Key Concepts for Quick Review:****

- Pipeline hazards and their solutions (forwarding, stalling, branch prediction)
- Cache organization types and their trade-offs
- Processor architecture differences (RISC vs CISC characteristics)

- Expression evaluation techniques (prefix, postfix, infix conversions)

****Calculation-Heavy Topics:****

- Cache hit ratio and effective access time calculations
- Pipeline performance and speedup calculations
- Floating-point representation and arithmetic
- Address mapping in different cache organizations

****Study Strategy**:** This topic requires understanding both theoretical concepts and practical implementations. Focus on how different architectural decisions impact performance and complexity.